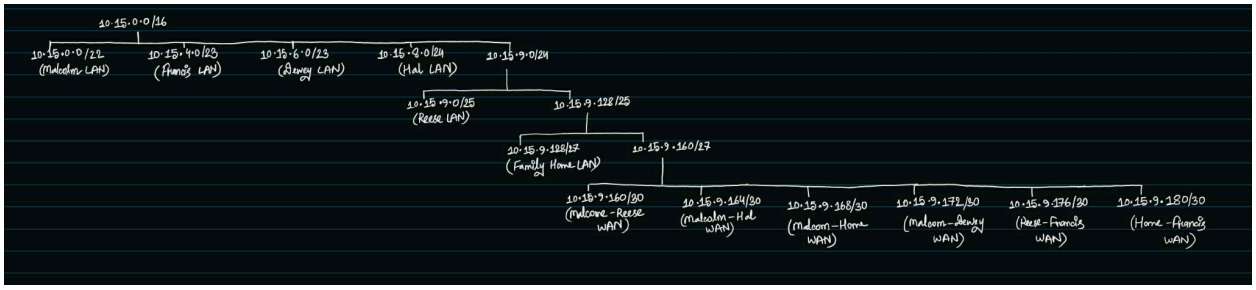
VLSM TREE:



Here we used VLSM to ensure minimal address waste at the subnet level.

Procedure of base network address derivation from my Student ID:

Student ID used: **24141215**

The last two digits of the Student ID were extracted and used as the second octet of a Class A private network in the format **10.X.0.0/16**

Step-by-step:

1. Take the 8-digit Student ID: 24141215
2. Use the last two digits: 15
3. Use 15 as the second octet of a /16 private network
4. Resulting base network: 10.15.0.0/16

The /16 prefix provides 65,534 usable host addresses, which comfortably accommodates the project's total host requirement of approximately 1,837 hosts (840 + 390 + 300 + 210 + 72 + 25) plus the inter-router point-to-point links, with sufficient room for future expansion. VLSM was then applied within this base to allocate appropriately-sized subnets to each zone, ensuring minimal address waste at the subnet level.

A brief note identifying everything:

Router-based DHCP:

1) Malcolm's School Lab: DHCP pool on R_Malcolm
2) Reese's Kitchen Control Zone: DHCP pool on R_Reese

Server-based DHCP:
1) Francis's Military Academy Network: DHCP_Francis server (10.15.4.2)
2) Hal's Workplace Office: DHCP_Hal server (10.15.9.2)
3) Dewey's Creative Experiment Lab: DHCP_Dewey server (10.15.6.3)

Static routing:
1) R_Francis: It uses standard static routes to all remote networks (including the required Francis ↔ Home standard static)
2) R_Dewey: uses standard static routes to all remote networks via R_Malcolm
3) R_Home: uses three static route types: a standard static route to Francis's network, a default static route toward Malcolm's network for all unknown traffic, and a floating static route to Dewey's network via Malcolm with administrative distance 25

Dynamic routing (RIPv2):
1) Malcolm's School Lab ↔ Reese's Kitchen Zone ↔ Hal's Workplace
2) Configured on R_Malcolm, R_Reese, and R_Hal with version 2, no auto-summary, network 10.0.0.0, and appropriate passive interfaces

The hybrid router is R_Malcolm:

1) Runs RIPv2 toward R_Reese and R_Hal (dynamic side)
2) Carries static routes toward R_Home (10.15.9.128/27) and R_Dewey (10.15.6.0/23) (static side)
3) Uses redistribute static under RIP so that R_Reese and R_Hal also learn the static destinations via RIP advertisements

Config files:

Malcolm

Hal

Reese

Francis

Home

Dewey

IP Address Table

1. Subnet Addresses and Masks

Zone	Subnet Address	Prefix	Subnet Mask	Usable Hosts	Host Range
Malcolm's School Lab	10.15.0.0	/22	255.255.252.0	1022	10.15.0.1 – 10.15.3.254
Francis's Military Academy	10.15.4.0	/23	255.255.254.0	510	10.15.4.1 – 10.15.5.254
Dewey's Creative Lab	10.15.6.0	/23	255.255.254.0	510	10.15.6.1 – 10.15.7.254
Reese's Kitchen Zone	10.15.8.0	/24	255.255.255.0	254	10.15.8.1 – 10.15.8.254
Hal's Workplace Office	10.15.9.0	/25	255.255.255.128	126	10.15.9.1 – 10.15.9.126
Family Home	10.15.9.128	/27	255.255.255.224	30	10.15.9.129 – 10.15.9.158

2. Point-to-Point Inter-Router Links (/30 each)

Link	Subnet	Mask	Router A IP	Router B IP
R_Malcolm ↔ R_Hal	10.15.9.160/30	255.255.255.252	10.15.9.161 (Malcolm, DCE)	10.15.9.162 (Hal)
R_Malcolm ↔ R_Reese	10.15.9.164/30	255.255.255.252	10.15.9.165 (Malcolm, DCE)	10.15.9.166 (Reese)
R_Malcolm ↔ R_Home	10.15.9.168/30	255.255.255.252	10.15.9.169 (Malcolm, DCE)	10.15.9.170 (Home)
R_Malcolm ↔ R_Dewey	10.15.9.172/30	255.255.255.252	10.15.9.173 (Malcolm, DCE)	10.15.9.174 (Dewey)
R_Reese ↔ R_Francis	10.15.9.176/30	255.255.255.252	10.15.9.177 (Reese, DCE)	10.15.9.178 (Francis)
R_Francis ↔ R_Home	10.15.9.180/30	255.255.255.252	10.15.9.181 (Francis, DCE)	10.15.9.182 (Home)

3. Default Gateways for Each Network Zone

Zone	Default Gateway	Gateway Router
Malcolm's School Lab	10.15.0.1	R_Malcolm
Francis's Military Academy	10.15.4.1	R_Francis
Dewey's Creative Lab	10.15.6.1	R_Dewey
Reese's Kitchen Zone	10.15.8.1	R_Reese
Hal's Workplace Office	10.15.9.1	R_Hal
Family Home	10.15.9.129	R_Home

4. Router Interface IPs

Router	Interface	IP Address	Subnet Mask	Connected Network
R_Malcolm	FastEthernet0/0	10.15.0.1	255.255.252.0	Malcolm LAN
	Serial0/1/0	10.15.9.161	255.255.255.252	Link to R_Hal
	Serial0/1/1	10.15.9.165	255.255.255.252	Link to R_Reese
	Serial0/3/0	10.15.9.169	255.255.255.252	Link to R_Home
	Serial0/3/1	10.15.9.173	255.255.255.252	Link to R_Dewey
R_Reese	FastEthernet0/0	10.15.8.1	255.255.255.0	Reese LAN
	Serial0/1/1	10.15.9.166	255.255.255.252	Link to R_Malcolm
	Serial0/1/0	10.15.9.177	255.255.255.252	Link to R_Francis
R_Hal	FastEthernet0/0	10.15.9.1	255.255.255.128	Hal LAN
	Serial0/1/0	10.15.9.162	255.255.255.252	Link to R_Malcolm
R_Francis	FastEthernet0/0	10.15.4.1	255.255.254.0	Francis LAN
	Serial0/0/0	10.15.9.178	255.255.255.252	Link to R_Reese
	Serial0/0/1	10.15.9.181	255.255.255.252	Link to R_Home
R_Dewey	FastEthernet0/0	10.15.6.1	255.255.254.0	Dewey LAN
	Serial0/0/0	10.15.9.174	255.255.255.252	Link to R_Malcolm
R_Home	FastEthernet0/0	10.15.9.129	255.255.255.224	Family Home LAN
	Serial0/0/0	10.15.9.170	255.255.255.252	Link to R_Malcolm
	Serial0/0/1	10.15.9.182	255.255.255.252	Link to R_Francis

5. Server IP Assignments (All Static)

Zone	Server	IP Address	Subnet Mask	Default Gateway	DNS Server
Family Home	Web_Server	10.15.9.130	255.255.255.224	10.15.9.129	10.15.9.131
Family Home	DNS_Server	10.15.9.131	255.255.255.224	10.15.9.129	10.15.9.131
Family Home	WilkersonHub	10.15.9.132	255.255.255.224	10.15.9.129	10.15.9.131
Malcolm Lab	Email_Malcolm	10.15.0.2	255.255.252.0	10.15.0.1	10.15.9.131
Dewey Lab	Email_Dewey	10.15.6.2	255.255.254.0	10.15.6.1	10.15.9.131
Francis Academy	DHCP_Francis	10.15.4.2	255.255.254.0	10.15.4.1	10.15.9.131
Hal Workplace	DHCP_Hal	10.15.9.2	255.255.255.128	10.15.9.1	10.15.9.131
Dewey Lab	DHCP_Dewey	10.15.6.3	255.255.254.0	10.15.6.1	10.15.9.131

6. Example End Device IP Assignments

Zone	Device	IP Address	Subnet Mask	Default Gateway	Assignment Method
Malcolm Lab	PC_Malcolm_1	10.15.0.11	255.255.252.0	10.15.0.1	Router-based DHCP
Malcolm Lab	PC_Malcolm_2	10.15.0.12	255.255.252.0	10.15.0.1	Router-based DHCP
Francis Academy	PC_Francis_1	10.15.4.12	255.255.254.0	10.15.4.1	Server-based DHCP
Francis Academy	PC_Francis_2	10.15.4.13	255.255.254.0	10.15.4.1	Server-based DHCP
Dewey Lab	PC_Dewey_1	10.15.6.13	255.255.254.0	10.15.6.1	Server-based DHCP
Dewey Lab	PC_Dewey_2	10.15.6.11	255.255.254.0	10.15.6.1	Server-based DHCP
Reese Zone	PC_Reese_1	10.15.8.11	255.255.255.0	10.15.8.1	Router-based DHCP
Reese Zone	PC_Reese_2	10.15.8.12	255.255.255.0	10.15.8.1	Router-based DHCP
Hal Workplace	PC_Hal_1	10.15.9.13	255.255.255.128	10.15.9.1	Server-based DHCP
Hal Workplace	PC_Hal_2	10.15.9.11	255.255.255.128	10.15.9.1	Server-based DHCP
Family Home	PC_Home_1	10.15.9.140	255.255.255.224	10.15.9.129	Static
Family Home	PC_Home_2	10.15.9.141	255.255.255.224	10.15.9.129	Static

7. DNS Records (on DNS_Server at 10.15.9.131)

Hostname	Type	IP Address	Purpose
www.family.net	A	10.15.9.130	Web Server
hub.family.net	A	10.15.9.132	WilkersonHub
dns.family.net	A	10.15.9.131	DNS Server itself
mail.malcolm.net	A	10.15.0.2	Malcolm's email server
mail.dewey.net	A	10.15.6.2	Dewey's email server
malcolm.net	A	10.15.0.2	Bare-domain (for SMTP relay)
dewey.net	A	10.15.6.2	Bare-domain (for SMTP relay)